

Building Autonomous Workflows with Sapphire

A practical description of the implemented agent API, its controls, and its limits.

This manual describes the current local implementation. Feature names are not evidence of production scale, hardware availability, or successful external actions.

Implemented loop

Agents retrieve bounded context, create a plan, select an action, check permissions, execute a registered or built-in action, record the observation, and report success or failure.

Memory

Working history and an in-process long-term key/value similarity store are available. SQLite FTS5 memory is provided by the industrial utilities. These are not hosted vector search or neural memory services.

Controls

Agent budgets cover steps, token budget metadata, elapsed time, and tool calls. Verifier and recovery callbacks can reject results or request another attempt.

Not included

Parser-level agent declarations, event triggers, DAG execution, unattended production daemons, and verified multi-agent recovery are not current language guarantees.

Version 1.0.5. Reproduce local measurements with `benchmarks/benchmark_runtime.py`. Values depend on the machine and environment.